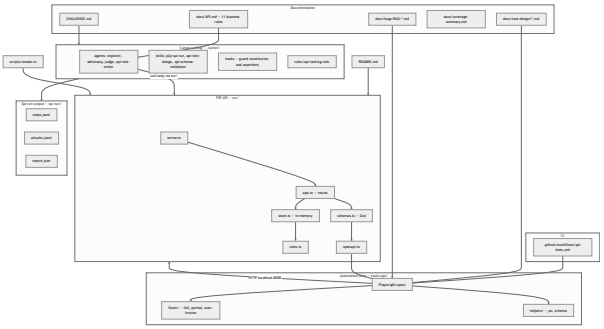
P2P API Practicum — project diagram

This document maps how the repository fits together: the practice API, documentation, automated tests, Cursor QA agents, and CI.

PDF export: run `npm run docs:diagram-pdf` to regenerate [project-diagram.pdf](./project-diagram.pdf) from this file.

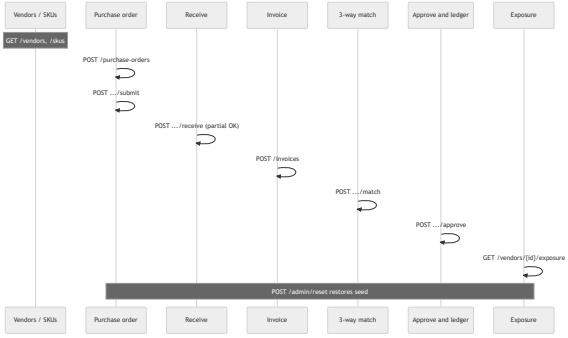
Repository overview



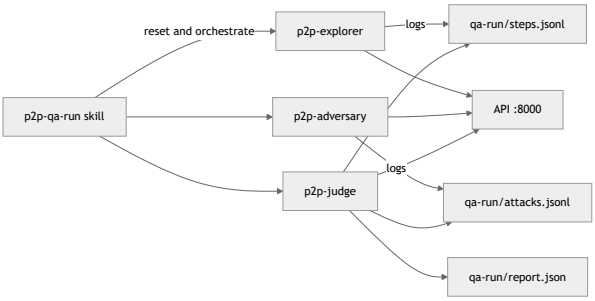
Runtime stack (request path)



Purchase-to-pay happy path



QA agent pipeline (challenge)



How the layers relate

Layer	Role
API	Express + Zod + in-memory store; intentional defects (find via HTTP, not by reading `src/`)
Docs	Human rules in `API.md`; machine schema via `/openapi.json` and Swagger at `/docs`
Playwright	Regression and flow tests under `tests/api/`; CI runs them on push
QA agents	Explorer, then adversary, then judge; artifacts under `qa-run/`
Bug reports	`docs/bugs/` documents findings tied to regression tests

Key paths

Path	Contents
`src/`	API implementation
`docs/API.md`	Written reference and money rules
`docs/test-design/`	Human-approved test design tables
`docs/bugs/`	Bug write-ups from test findings
`tests/api/`	Playwright API tests
`.cursor/agents/`	Subagent prompts (explorer, adversary, judge, test writer)
`.cursor/skills/p2p-qa-run/`	Orchestrates the QA agent run
`qa-run/`	Output from the last QA agent run
`CHALLENGE.md`	60-minute QA agent exercise spec